

MetrCheck AI

AI-Powered Legal Metrology Compliance Assistant | SIH Problem Statement: SIH26034

1. Executive Summary & Objective

MetrCheck AI is an intelligent, automated compliance auditing platform engineered to inspect packaged commodity labels for strict conformity with the **Legal Metrology (Packaged Commodities) Rules, 2011** and **FSSAI Food Safety and Standards (Packaging & Labelling) Regulations**. By pairing deep learning Optical Character Recognition (OCR) with deterministic statutory rule matrices and structured heuristic extraction, MetrCheck AI automatically discovers mandatory declarations, detects legal defects/omissions, highlights physical bounding box evidence, and generates audit-grade compliance reports.

2. System Technology Stack & Role Matrix

Layer / Component	Technology Used	Role & Functional Purpose
Frontend UI	React 19, TypeScript, Vite, Tailwind CSS v4	High-performance responsive single-page web app with Light/Dark mode and multi-slot panel uploading.
Backend Framework	FastAPI (Python 3.11), Uvicorn, Pydantic v2	Asynchronous high-throughput REST API with strict request/response data schema validation.
Primary OCR Engine	PaddleOCR (PP-OCRv4: DBNet + SVTR)	Deep learning detection, angle classification, text recognition, and spatial 2D bounding box coordinate mapping.
Fallback OCR	Tesseract OCR (pytesseract)	Secondary multi-pass OCR fallback for offline systems lacking deep learning hardware.
Computer Vision / Quality	OpenCV (cv2) & Pillow (PIL)	Image quality assessment (Laplacian blur variance, brightness, contrast) & image normalization.
Extraction Engine	Compiled Regex + Heuristics + Typo Repair	Identifies statutory entities (MRP, Net Qty, Dates, FSSAI Lic No, Address, Consumer Care).
Compliance Engine	Declarative Python Rule Matrix (13 Rules)	Codifies 8 Legal Metrology + 5 FSSAI rules into deterministic compliance verdicts.
Database Layer	SQLite (aiosqlite)	Asynchronous persistent storage for scan history, audit trails, and dashboard statistics.
Data Visualization	Recharts & Lucide Icons	Compliance score gauge circles, historical trend charts, and clear iconography.
Report Generation	ReportLab & Jinja2 Templates	Automated generation of downloadable compliance audit PDF and HTML summary dossiers.

3. Step-by-Step Pipeline & Working Architecture

Workflow: Image Ingestion → Quality Assessment → OCR Processing → Entity Extraction → Rule Verification → Scoring → Report Dossier

- Step 1: Multi-Panel Ingestion:** Accepts packaging artwork across 4 dedicated slots: *Front*, *Back*, *Side 1*, *Side 2* in JPG/PNG/WebP formats up to 10 MB.

- **Step 2: Preprocessing & Quality Screening:** OpenCV computes Laplacian variance to detect camera blur (< 70 threshold), checks brightness (< 45 underexposed, > 230 glare), and verifies resolution (> 400px).
- **Step 3: Deep Learning OCR & Bounding Boxes:** PP-OCRv4 extracts raw text fragments while tracking exact spatial [xmin, ymin, xmax, ymax] bounding boxes. Packaging typo-repair fixes character confusions (e.g., '400 9' → '400 g', 'Lic No' optical noise).
- **Step 4: Information & Attribute Extraction:** Robust pattern matching isolates critical fields: MRP, standard metric Net Quantity, Manufacturing/Packaging Dates, Expiry/Best Before, FSSAI 14-digit Licence, Manufacturer & PIN Code, Country of Origin, Customer Helpline.
- **Step 5: Statutory Rules Engine:** Evaluates extracted declarations against 13 codified rules. Determines status: **PASS**, **WARNING**, **NEEDS_REVIEW**, **FAIL**, or **NOT_APPLICABLE**.
- **Step 6: Explainable Scoring & Prioritized Actions:** Calculates a deterministic 0–100 score. Generates statutory clause references and actionable corrective steps for immediate remediation.

4. Codified Statutory Compliance Rules Matrix

The system enforces 13 specific statutory rules across two regulatory domains:

Rule ID	Statutory Reference	Rule Title & Requirement	Severity
LM-001	Rule 6(1)(a)	Manufacturer / Packer / Importer: Name & full physical address with PIN code.	HIGH
LM-002	Rule 6(1)(b)	Generic / Common Name: Specific commodity name on Principal Display Panel.	HIGH
LM-003	Rule 6(1)(c) & 12	Net Quantity: Standard metric unit (g, kg, ml, L) & minimum font height standards.	HIGH
LM-004	Rule 6(1)(e)	Maximum Retail Price (MRP): Formatted in Rs/■ with 'inclusive of all taxes'.	HIGH
LM-005	Rule 6(1)(f)	Consumer Care Details: Name, address, phone helpline number, and email.	HIGH
LM-006	Rule 6(10)	Country of Origin: Mandatory declaration for both imported & domestic goods.	MEDIUM
LM-007	Rule 6(11)	Unit Sale Price (USP): Price per gram/milliliter for retail commodities.	MEDIUM
LM-008	Rule 6(1)(d)	Date of Manufacture / Packing: Conspicuous month & year (MM/YYYY) marking.	HIGH
FS-001	Regulation 5(4)	FSSAI Logo & Licence: Valid 14-digit numeric FSSAI registration number.	HIGH
FS-002	Regulation 5(1)	Name of Food: Standard food category & true nature identification.	HIGH
FS-003	Regulation 5(2)	Ingredients List: Descending order list & mandatory allergen declaration.	MEDIUM
FS-004	Regulation 5(3)	Nutritional Information: Energy, Protein, Carb, Fat per 100g/100ml.	HIGH
FS-005	Regulation 5(7)	Date Marking: 'Best Before' or 'Expiry Date' with storage instructions.	HIGH

5. Compliance Scoring Algorithm & Classification

The scoring algorithm evaluates all applicable rules using configured statutory screening weights, strictly excluding non-applicable regulations (e.g. non-food products are not penalized for lacking FSSAI rules):

Formula: $\text{Score} = (1.0 \times \text{PASS} + 0.85 \times \text{REVIEW} + 0.5 \times \text{WARNING} + 0.0 \times \text{FAIL}) / (\text{Total Applicable Rules}) \times 100$

- **COMPLIANT (Score ≥ 85%):** All mandatory statutory declarations are present, valid, and fully verified.
- **POTENTIAL NON-COMPLIANCE (Has FAIL):** One or more confirmed statutory failures (e.g., missing MRP, missing manufacturer address, or missing FSSAI licence).
- **REVIEW REQUIRED (Has Review items):** Declarations detected with moderate confidence or non-critical formatting ambiguities requiring manual inspector verification.

6. Pre-Configured Benchmark Demo Cases

- **Demo Case 1 (Compliant):** Royal Gold Premium Basmati Rice (Score ~95.5%) — Complete declarations across front & back panels.
- **Demo Case 2 (Warning / Review):** FreshVita Mixed Fruit Juice (Score ~76.9%) — Missing consumer care email & non-standard date format.
- **Demo Case 3 (Violation):** QuickBite Instant Noodles (Score ~37.9%) — Multiple severe omissions (Missing MRP, Address, Net Weight, Dates).

7. Local Deployment & Running Guide

- **Backend API Server:** <http://localhost:8000> (FastAPI + Uvicorn)
- **Frontend Dashboard:** <http://localhost:5173> (React + Vite)
- **Interactive OpenAPI Docs:** <http://localhost:8000/docs>
- **One-Click Launch:** Run `run.bat` to start both servers concurrently.